

Logic Gates — Paper C

HARD • NON-CALCULATOR • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. The mark for each question is in square brackets on the right.
- No calculator is needed anywhere on this paper.
- Truth-table rows go in **binary counting order**: 00, 01, 10, 11, and for three inputs 000, 001, 010, 011, 100, 101, 110, 111.
- Write expressions with AND, OR, NOT, NAND and NOR, and put **brackets wherever two different gates meet**. NOT applies only to the thing straight after it, so NOT A AND B means (NOT A) AND B.
- Several questions describe machines you may not have seen before. You are not expected to recognise them: apply the gate rules and the one-gate-at-a-time method you already have.
- No reminder of the gates is printed on this paper.

SECTION 1 – QUICK AND EXACT

- 1.** Each output column below belongs to a **three-input** AND, OR, NAND or NOR gate, and each gate is used once. Write the name of the gate in the bottom row. [1]
- One mark for each correct pair.

A	B	C	(i)	(ii)	(iii)	(iv)
0	0	0	1	0	1	0
0	0	1	1	0	0	1
0	1	0	1	0	0	1
0	1	1	1	0	0	1
1	0	0	1	0	0	1
1	0	1	1	0	0	1
1	1	0	1	0	0	1
1	1	1	0	1	0	1
Gate						

- 2.** Answer these without drawing the tables.

(a) A truth table has **64** rows. How many inputs does the circuit have? [1]

(b) A circuit has four inputs, A, B, C and D. How many of its rows have $A = 1$? [1]

(c) A **four-input NAND** gate has 16 rows. On how many of them is the output 1? [1]

3. In the first expression NOT applies only to A. Complete the table for both:

$$X = \text{NOT } A \text{ AND } B \text{ AND } C \quad Y = \text{NOT } (A \text{ AND } B \text{ AND } C)$$

- (a) Complete the table.

[2]

A	B	C	X	Y
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

- (b) On how many of the 8 rows do X and Y give different outputs?

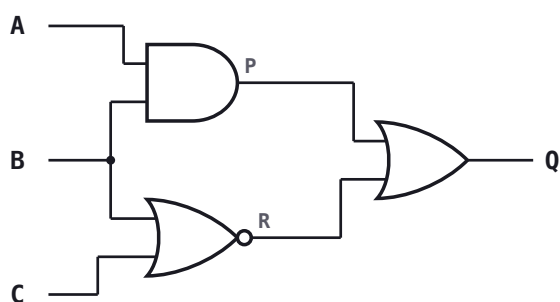
[1]

- (c) Y gives the same table as one single three-input gate. Name it.

[1]

SECTION 2 – CIRCUITS AND TABLES

4. Study the circuit. P and R are the outputs of the first two gates.



- (a) Write the expression for Q.

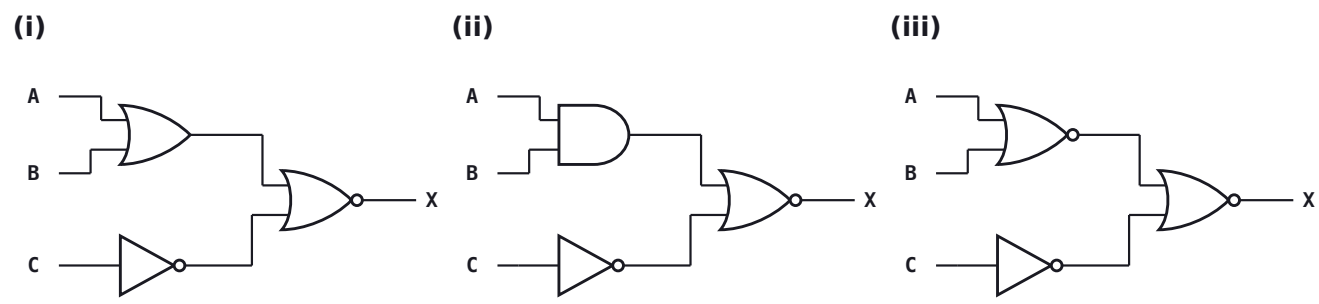
[2]

- (b) Complete the trace table.

[3]

A	B	C	P	R	Q
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

5. The table gives the output X of exactly one of the three circuits below. The circuits are identical except for one gate, so they agree on most rows. Do not complete all three tables.



A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

(a) Tick the circuit that produces X. [1]

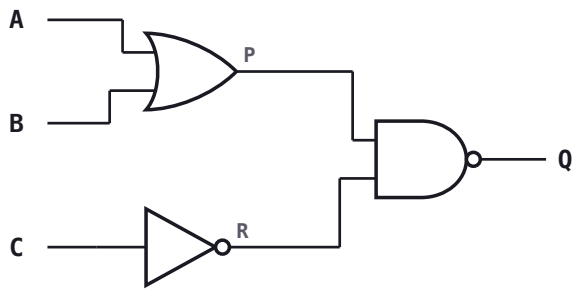
Circuit (i) ☐

Circuit (ii) ☐

Circuit (iii) ☐

(b) Write down one row (as three digits, ABC) that rules out circuit (i), and one row that rules out circuit (iii). Say what each wrong circuit outputs on its row. [2]

6. Tom filled in the trace table for the circuit below. He made one mistake, in column P, and carried it forward.



A	B	C	P	R	Q
0	0	0	0	1	1
0	0	1	0	0	1
0	1	0	1	1	0
0	1	1	1	0	1
1	0	0	1	1	0
1	0	1	1	0	1
1	1	0	0	1	1
1	1	1	0	0	1

- (a) Write down the rows (ABC) where column P is wrong. [1]

- (b) Describe the rule Tom used for gate P that is not the rule for that gate. [1]

- (c) How many cells in column Q are wrong? [1]

SECTION 3 – MACHINES AND DESIGN

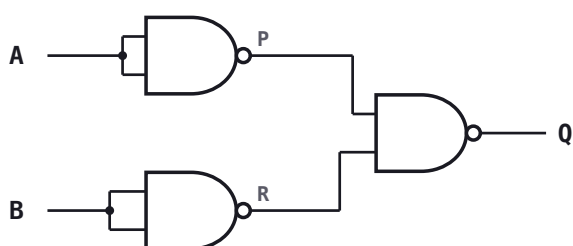
7. A greenhouse in Nashik has a misting valve, Q. **A = 1** when the air is too hot. **B = 1** when the soil is wet. **C = 1** when the gardener presses the manual button. **D = 1** when the water pump is running. The valve opens when (the air is too hot and the soil is **not** wet, or the manual button is pressed) **and** the pump is running. With the pump off, the valve must stay shut whatever else is happening.

- (a) Write an expression for Q. [2]

- (b) The full table has 16 rows. Without writing it out, work out how many of them give $Q = 1$. Show your reasoning. [2]

- (c) State Q when $A = 1$, $B = 1$, $C = 0$ and $D = 1$, and give the reason in a few words. [1]

8. A workshop has only **NAND** gates in stock. A technician builds the circuit below, in which P and R are the outputs of the first two gates.



- (a) Complete the trace table. [2]

A	B	P	R	Q
0	0			
0	1			
1	0			
1	1			

- (b) Explain why the first two gates act as NOT gates, even though they are NAND gates. [1]

- (c) Draw one NAND gate, wired to behave as a NOT gate, in the box. [1]



- (d) State the single gate that the circuit above is equivalent to. [1]

End of paper. Check every truth table has all its rows in counting order, and that every expression has brackets wherever two different gates meet.